

Cholesky:

$$\begin{cases} 4x_1 + 2x_2 + 6x_3 = -2 \\ 2x_1 + 2x_2 + 5x_3 = 0 \\ 6x_1 + 5x_2 + 22x_3 = 1 \end{cases}$$

define one-point iterative method to approximate the root $\sqrt{5}$, for a sufficiently close initial value x_0 . Find the order of convergence, bound for the error of approx.

 $f \in C^3[0,1]$
a) $(P_f)(x)$, knowing $f(0), f'(0)$ and $f(1)$ b) Integrate $(P_f)(x)$ to obtain a quadrature formula for $\int_0^1 f(x) dx$. Find its degree of precision.c) Find A_0, A_1, A_2 s.t. the quadrature formula

$$\int_0^1 f(x) dx = A_0 f(0) + A_1 f'(0) + A_2 f(1) + R(f)$$

has the max. degree of precision possible. Compare the formula to the one in part b)

d) Find the remainder $R(f)$

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$$A = \left[\begin{array}{c|cc} 4 & 2 & 6 \\ \hline 2 & 2 & 5 \\ 6 & 5 & 22 \end{array} \right] \sim \left[\begin{array}{c|cc} \sqrt{4} & & \\ \hline 2/2 & & \\ 6/2 & & \end{array} \right]$$

$$\begin{aligned} A' - w w^T / a_{11} &= \begin{bmatrix} 2 & 5 \\ 5 & 22 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 2 & 6 \end{bmatrix} / 4 \\ &= \begin{bmatrix} 2 & 5 \\ 5 & 22 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ 2 & 13 \end{bmatrix} \Rightarrow \left[\begin{array}{c|c} 2 & \\ \hline 1 & \sqrt{1} \\ 3 & 2 \end{array} \right] \sim \begin{bmatrix} 2 & \\ 1 & \sqrt{1} \\ 3 & 2\sqrt{1} \end{bmatrix} \end{aligned}$$

$$A' - w \cdot w^T / a_{11} \Rightarrow [13] - [2] \cdot [2] / 1 \Rightarrow [13] - [4] = [9]$$

$$R^T \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 2 & 3 \end{bmatrix} \Rightarrow R = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

$$A = R^T \cdot Q = L \cdot U$$

$$L y = b$$

$$U x = y$$